



Zero-birefringence dual mode waveguides and polarization-independent two-mode (de)multiplexer on thin film Lithium niobate

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ABSTRACT

Zero birefringence (ZBR) dual-mode rib waveguides supporting two quasi TE modes (TE_{00} , TE_{10}) and two quasi TM modes (TM_{00} , TM_{10}) are designed on thin film X-cut lithium niobate. Anisotropy of lithium niobate is utilized to achieve a birefringence of $\sim 2 \times 10^{-5}$ for both fundamental mode pair and the first order mode pair simultaneously, using full-vectorial finite difference Eigen mode solver. Coherence length of 1.4 mm is achieved with practically acceptable tolerances to fabrication errors. ZBR waveguides have been used to design polarization-independent two-mode (de) multiplexer with mode conversion efficiency greater than 96% and a crosstalk of about -14 dB for both TE and TM polarizations. Proposed device is tolerant to fabrication errors and exhibits a broad bandwidth of 165 nm. Polarization-independent components reduce system complexity in mode/polarization division hybrid multiplexing schemes and are also useful in multi-analyte sensing.

1. Introduction

High index contrast photonic platforms such as silicon on insulator (SOI) and lithium niobate on insulator (LNOI) exhibit strong birefringence, where the effective indices of transverse-electric and transverse magnetic modes are considerably different [1]. This results in polarization-sensitive behavior of the photonic devices, which necessitates polarization diversity schemes. An alternative approach is to design polarization-independent or zero-birefringence waveguides that have same effective index for both quasi transverse electric (qTE) and quasi transverse magnetic (qTM) modes, so that they have same phase along the propagation. Such devices reduce the system complexity by eliminating the need for polarization-diversity schemes that process qTE and qTM modes separately [2]. State of the art techniques to achieve zero birefringence (ZBR) include geometry control of rib waveguides [3], material index variation in slot waveguides [4], subwavelength structures [5], and stress engineering in waveguide clad layers [6]. All these techniques are reported for only fundamental qTE and qTM modes of single mode waveguides. However, zero birefringence is desired for multimode waveguides as well in hybrid multiplexing schemes incorporating mode division multiplexing (MDM) and polarization division multiplexing (PDM) [7] to mitigate the phase dependent losses and dispersion that can have negative influence affecting efficiencies of MDM devices [8,9]. Such hybrid scheme doubles the capacity of MDM link without additional components like polarization beam splitters. ZBR in multimode waveguides is also useful for multi-analyte sensing [10].

In this work, we report a dual-mode rib waveguide on X-cut thin film lithium niobate platform in which, zero birefringence is achieved using the interplay of waveguide birefringence and intrinsic birefringence of lithium niobate (LN). Using ZBR waveguides, we have also designed a broadband polarization-independent two-mode multiplexer/demultiplexer (mux/demux). With the advent of thin film lithium niobate (~ 400 – 800 nm thick) [11] similar to SOI, there has been a renewed world-wide interest in considering this material for photonics. The optical mode size in LNOI waveguides is typically reduced by more than an order of magnitude compared to their bulk counterparts [12], which favors compact photonic devices. LNOI is an excellent platform for integrated photonic devices because of its remarkable properties like broad transparency range from 350 nm to 5200 nm, strong electro-optic, acousto-optic, and thermo-optic effects [13].

LN is a negative uniaxial material with ordinary index (n_{or}) along two of the principal axes and, extraordinary index (n_{ex}) along the optic axis of the crystal. We have utilized this anisotropy to tune the modal birefringence in dual mode rib waveguide supporting two qTE and two qTM modes. A low birefringence of 2.5×10^{-4} is reported in single mode buried LN waveguides [9], where qTE modes are attributed to extraordinary and qTM modes to ordinary polarizations respectively. While [9] reports polarization independence for only fundamental modes, we have achieved a birefringence which is almost an order of magnitude less (2×10^{-5}) for both fundamental mode pair and higher order mode pair. Such ZBR waveguides facilitate the design of

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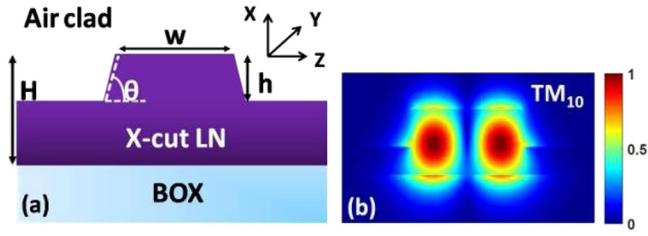


Fig. 1. Lithium niobate waveguide (a) cross-section (b) mode profile of TM_{10} mode.

polarization-independent components. One such device namely, two-mode polarization-independent mux/demux has also been designed and analyzed in this work.

Different approaches have been proposed to build polarization-insensitive components like MMI based power splitter [14], material index variation in slot waveguide [15], and subwavelength structures [16] on SOI platform. However, there have been a very few such attempts in LNOI platform. In fact, it is easier to achieve ZBR in LN as it offers additional degree of freedom in the form of intrinsic birefringence. Polarization-independent coupler proposed in [17], utilizes orientation engineering in LN. This work suggests off-axis propagation in LN waveguides, which is not a good choice while implementing electro-optic effect. All the above mentioned solutions are restricted to single mode waveguides. To the best of our knowledge, this work is the first attempt towards building polarization-insensitive multimode devices.

2. Methodology

The waveguide cross-section considered for the design of zero birefringence waveguides along with the crystal principal axes is shown in Fig. 1(a) and mode profile of TM_{10} in Fig. 1(b). Only waveguide dimensions are varied and there is no rotation of the principal axes which retains the permittivity tensor diagonal [18].

In this configuration, the waveguide is aligned along the Y-direction which can achieve strongest electro-optic effect using the highest electro-optic coefficient r_{33} of LN, when the applied electric field is aligned along the extraordinary axis Z [19].

Proposed rib waveguide has a film thickness (H) of 700 nm with an etch depth (h) of 400 nm and a sidewall angle (θ) of 83.5° with an underlying buried oxide layer of 2 μm . Even though LN is a difficult material to etch, sidewall angle of 83° has already been reported [11]. Rib configuration is chosen as the effective indices of adjacent modes are spaced closest [20] and hence offer a higher chance of achieving ZBR. Waveguide dimensions are chosen to have more than 85% power confinement for all the four guided modes. TE polarization fraction of qTE modes is 0.99 and that of qTM modes is 0.03, thus avoiding hybrid modes completely. Hybrid modes result in large group velocity dispersion [21,22] and can even lead to unwanted coupling between modes [8]. These ZBR waveguides are used to design polarization-insensitive two-mode mux/demux using asymmetric directional coupler. Broadband mode-division mux/demux for TE polarization have been reported in our previous work [23], while current work reports polarization-independent multimode mux/demux.

3. Results and discussion on ZBR waveguides

The evolution of effective indices of guided modes in the designed rib waveguide is shown in Fig. 2(a). Numerical analysis is carried out at a wavelength of 1550 nm with $n_{\text{or}} = 2.21116$, $n_{\text{ex}} = 2.13755$ for LN, and $n = 1.4433$ for the box layer.

It is interesting to note that the effective indices of fundamental modes TE_{00} and TM_{00} follow each other very closely over a width of 1.65 μm to 1.85 μm . Even more interesting is the fact that, effective

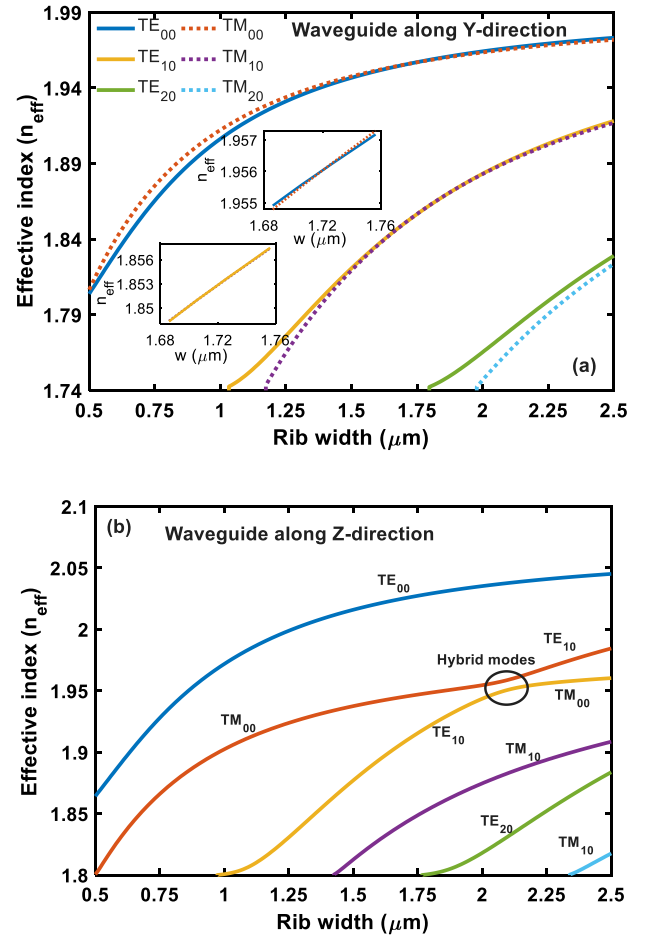


Fig. 2. Effective indices of guided modes (a) Y-propagating (b) Z-propagating LN with the rib width variation at a fixed etch depth of 400 nm and $H = 700$ nm. Effective indices of TE_{00} and TM_{00} (as well as TE_{10} and TM_{10}) are almost equal over $w = 1.65$ to 1.85 μm in Y-propagating LN.

indices of next higher order mode pair TE_{10} and TM_{10} also track each other very closely over the same width range. Therefore, when the rib width is chosen in this range, it is possible to obtain a dual mode waveguide with zero birefringence. It can also be observed that there are no anti-crossings in the curves, indicating that there are no hybrid modes. The behavior of modes in a waveguide with same dimensions but propagation direction aligned along Z-axis is shown in Fig. 2(b). This configuration is equivalent to isotropic material and waveguide birefringence dominates resulting in a strongly polarization-dependent waveguide. Anti-crossing is also observed at a $w = 2.08 \mu\text{m}$ resulting in mode hybridization between TM_{00} and TE_{10} .

Effective index difference between the mode pairs is taken as the indicative measure of modal birefringence, represented as $|\Delta n_0| = |n_{\text{eff}TE_{00}} - n_{\text{eff}TM_{00}}|$ and $|\Delta n_1| = |n_{\text{eff}TE_{10}} - n_{\text{eff}TM_{10}}|$. For the Y-propagating waveguide in Fig. 2(a), $|\Delta n_0| = 0.2 \times 10^{-5}$ and $|\Delta n_1| = 2 \times 10^{-5}$ respectively when rib width is 1.72 μm . Such small birefringence values translate to a large coherence length of 38 mm when operated at a wavelength of 1550 nm. This value meets the typical device length requirement needed in integrated photonics. However, fabrication tolerances to width, sidewall angle and etch depth need to be investigated as shown in Fig. 3 in order to find practically achievable coherence lengths. Fabrication error ± 100 nm in rib width alters $|\Delta n_0|$ and $|\Delta n_1|$ by just 3.5×10^{-4} as shown in Fig. 3(a). An etch depth variation of ± 10 nm still retains both the birefringences below 1.2×10^{-3} . It can be observed from Fig. 3(b) that $|\Delta n_1|$ is very sensitive to changes in etch depth compared to all other design

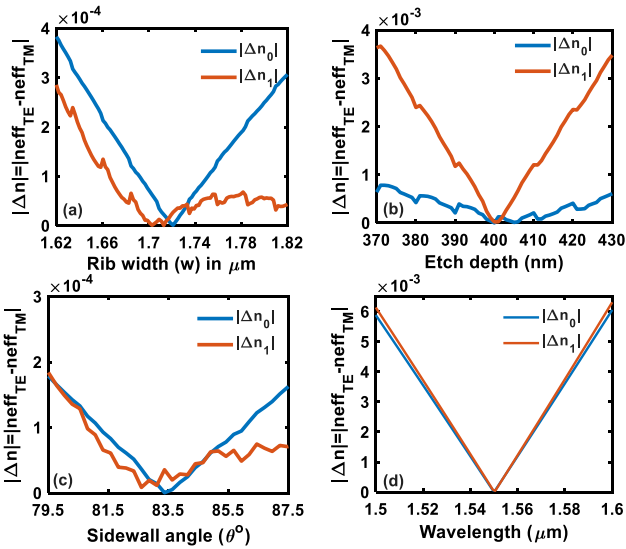


Fig. 3. Tolerance to (a) rib width (b) etch depth (c) sidewall angle variation (d) wavelength. Width deviation of ± 50 nm has negligible effect on birefringence. A sidewall angle deviation of $\pm 4^\circ$ causes a small change of 1.6×10^{-4} in $|\Delta n|$.

parameters. Sidewall angle has a negligible influence on Δn , causing a change 1.6×10^{-4} for a deviation of $\pm 4^\circ$. This dual mode waveguide exhibits a moderate spectral width of 80 nm for Δn to be maintained below 0.005 as shown in Fig. 3(d). Wavelength dependent refractive index of LN is modeled using Sellmeier equation [24].

In what follows, modal birefringence control using anisotropy of LN is extended to several other designs for different sidewall angles of 75° to 90° (ideal vertical sidewalls), and etch depths of 400 nm to 500 nm over an appropriate range of rib widths. All the numerical analyses are done with a film thickness of 700 nm at $\lambda = 1550$ nm, using Lumerical's full vectorial finite difference Eigen mode solver.

LN is known to be a difficult material to etch and generally result in angled sidewalls [13]. For the first set of plots, we consider sidewall angles of 75° to 90° with 'h' at 400 nm, 'w' varied from 1.22–1.75 μm . The range of width is chosen to guide only four modes TE_{00} , TM_{00} , TE_{10} , and TM_{10} . Contours maps with different $|\Delta n|$ starting from the least 2×10^{-5} up to 2×10^{-3} are plotted. White area to the left in Fig. 4(b) indicate the regions where $|\Delta n_1| > 5 \times 10^{-3}$. It is observed that for any value of the sidewall angle from 75° to 90° , there is a range of widths to choose from, so as to minimize $|\Delta n|$.

For the next set of plots, etch depth is varied from 300 nm to 500 nm along with the width as shown in Fig. 5. The sidewall angle is fixed at 83.5° . However, the designer is free to choose any sidewall angle over a range of 75° to 90° and still achieve desired ZBR as inferred from Fig. 4. Sample contours for which $|\Delta n|$ are in the range of 2×10^{-5} to 5×10^{-3} are plotted to avoid crowded plots for better readability. Dotted, dashed and solid curves indicate $|\Delta n| \leq 1 \times 10^{-4}$, $|\Delta n| < 1 \times 10^{-3}$, and $|\Delta n| \leq 2 \times 10^{-3}$ respectively. At larger widths, the influence of etch depth on modal birefringence is smaller.

It is possible to achieve $|\Delta n_0| < 1 \times 10^{-3}$ for etch depths of 335–460 nm with a proper choice of rib width. However, $|\Delta n_1|$ being more sensitive to etch depth, limits usable values to 387–420 nm for the same $|\Delta n|$. $|\Delta n_0|$ hits a zero at $w = 1.72 \mu\text{m}$ while $|\Delta n_1|$ hits zero at $w = 1.713 \mu\text{m}$ respectively at an etch depth of 400 nm. So the best working point is around an etch depth of 400 nm with the width varied around 1.7 μm . Regions where the desired modes are cutoff ($N < 4$) or additional modes are guided ($N > 4$) are shown as white regions in Fig. 5.

To the best of our knowledge, ZBR has only been achieved in single-mode waveguides so far and this work reports polarization independence in dual-mode waveguides. Therefore, it is difficult to compare the results with the existing literature. However, an attempt has been

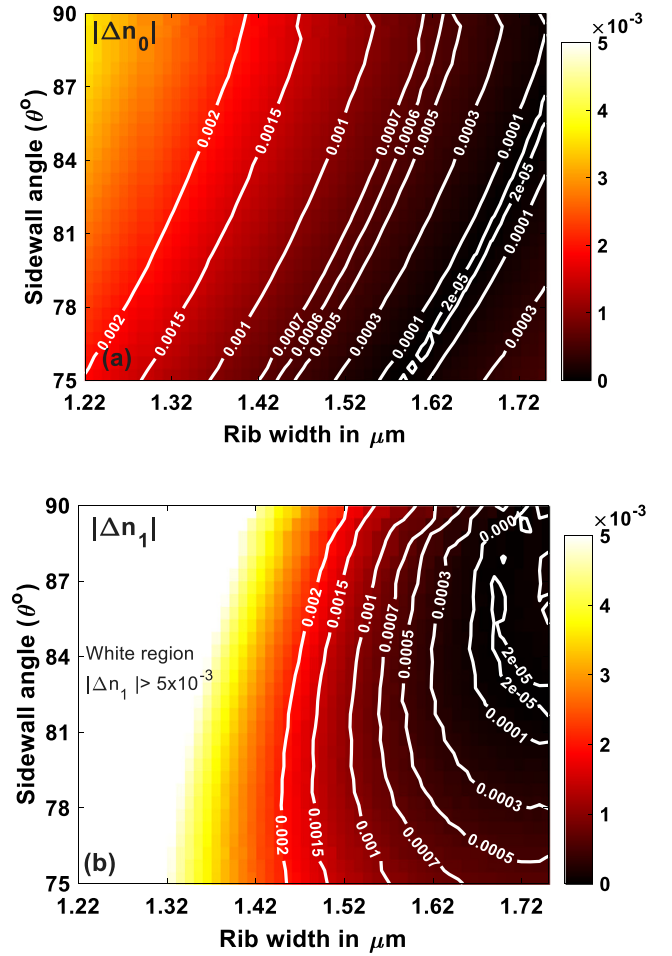


Fig. 4. 2D plot of birefringence (a) $|\Delta n_0|$ (b) $|\Delta n_1|$ as a function of rib width and sidewall angle. At any sidewall angle from 75° to 90° , rib width can be chosen to restrict $|\Delta n| < 5 \times 10^{-4}$.

made in Table 1 to summarize the existing ZBR waveguides along with the results of current work. [4] and [5] are ZBR waveguides on silicon, while [9] and this work explore polarization independence in lithium niobate platform. Proposed design has achieved the lowest birefringence for not just the fundamental mode pair but also for the higher order mode pair. Also, this structure is simple compared to slot and sub-wavelength structures proposed in [4] and [5].

Larger dimensions of LN waveguide also allow a large fabrication tolerance compared to the existing designs. If the maximum allowed birefringence is 3.5×10^{-3} , then the etch depth tolerance can increase to ± 30 nm.

4. Design of polarization-independent two-mode mux/demux

The (de)mux consists of a narrow waveguide (wg1) supporting fundamental TE mode along with fundamental TM mode, and a wide waveguide (wg2) supporting TE_{00} , TE_{10} as well as TM_{00} , TM_{10} separated by a gap (g) as shown in Fig. 6. The device structure is based on asymmetric directional coupler similar to that proposed in [23]. When TM_{00} (TE_{00}) is launched into port-1, mode TM_{10} (TE_{10}) emerges out of port-4 at the wide waveguide, as a result of phase matching between these modes.

For the mux/demux to be polarization independent, the coupling lengths (L_c^{TE} and L_c^{TM}) defined in Eqs. (1) and (2) for TE and TM modes have to be equal. One way of achieving this is to use dual-mode ZBR waveguides to build the asymmetric directional coupler. It

Table 1
Comparison of ZBR waveguides in terms of birefringence, tolerance and bandwidth.

Ref.	Principle	N ^a	Δn	BW (nm) for Δn < 5 × 10 ⁻³	Tolerance in nm (Δn < 1 × 10 ⁻³)	
					Δw	Δh
[5]	Sub-λ structures	1	0	90	5	2
[4]	Slot	1	5 × 10 ⁻⁴	–	3	–
[9]	Anisotropy of LN	1	2.5 × 10 ⁻⁴	200	6	3.5
This work	Anisotropy of LN	2 ^b	0.2 × 10 ⁻⁵ ^c 2 × 10 ⁻⁵	80	400	8

^aN- number of modes in the ZBR waveguide.

^b2 TE modes and 2 TM modes.

^c|Δn| between TE₁₀ and TM₁₀.

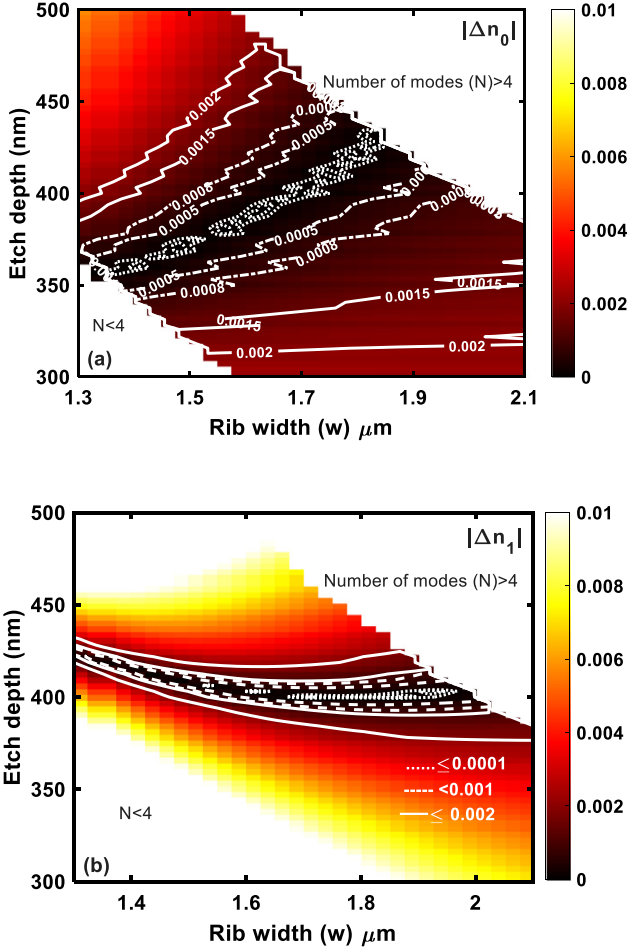


Fig. 5. 2D plot of birefringence (a) $|\Delta n_0|$ (b) $|\Delta n_1|$ as a function of rib width and etch depth. $|\Delta n_1|$ is more sensitive to etch depth and limits the usable 'h' values to achieve zero birefringence.

is observed from Fig. 2a that the fundamental modes of orthogonal polarization (TE₀₀ and TM₀₀) have almost same effective indices when the width of the waveguide is chosen in the range of 1.5–1.8 μm. Also, first order modes (TE₁₀ and TM₁₀) have equal effective indices making the dual mode waveguide polarization insensitive. In order to achieve phase matching between TE₀₀ mode of narrow waveguide and TE₁₀ mode of wide waveguide, width ' w_1 ' of narrow waveguide is chosen to be 0.6 μm, width ' w_2 ' of wide waveguide as 1.595 μm, and the gap between them as 250 nm. With this choice of widths at an etch depth of 400 nm on a film of thickness 700 nm, L_c^{TE} and L_c^{TM} were found to be 23.79 and 23.53 μm respectively. This small difference of 0.26 μm in the coupling lengths does not affect the performance of the device. More stringent limitation is the accuracy of phase matching the modes,

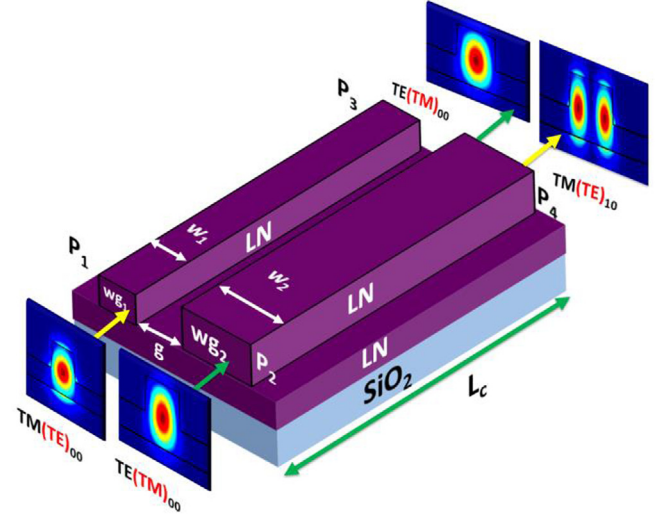


Fig. 6. Schematic of polarization-independent two-mode mux/demux.

between which coupling is needed. For the proposed mux/demux, average of the two coupling lengths is taken, which is 23.66 μm.

$$L_c^{TE} = \frac{\lambda}{2(nTE_{even} - nTE_{odd})} = \frac{\lambda}{2(\Delta n_{TE})} \quad (1)$$

$$L_c^{TM} = \frac{\lambda}{2(nTM_{even} - nTM_{odd})} = \frac{\lambda}{2(\Delta n_{TM})} \quad (2)$$

where, λ is the operating wavelength, nTM_{even} and nTM_{odd} are the effective indices of even and odd supermodes of asymmetric directional coupler [25].

Apart from the widths of individual waveguides, L_c is a strong function of the separation between the two waveguides as shown in Fig. 7(a), where the two coupling lengths become equal at a gap of 255 nm. Considering fabrication difficulties, g is chosen as 250 nm for rest of the analysis. Fig. 7(b) and (c) show the evolution of dominant field components of TM (E_y) and TE (E_x) modes along the propagation direction. This configuration achieves a mode conversion efficiency (MCE) >96% and a crosstalk (CT) ~ -14 dB for both TE and TM polarizations.

Mode conversion efficiency and crosstalk [26] are defined in (3) and (4).

$$MCE^{TE} = \frac{P_{41}^{TE}}{P_{41}^{TE} + P_3}, \quad MCE^{TM} = \frac{P_{41}^{TM}}{P_{41}^{TM} + P_3} \quad (3)$$

$$CT^{TE} = 10 * \log_{10} \frac{P_{30}^{TE}}{P_{41}^{TE}}, \quad CT^{TM} = 10 * \log_{10} \frac{P_{30}^{TM}}{P_{41}^{TM}} \quad (4)$$

where, $P_{41}^{TE(TM)}$ is power in TE₁₀ (TM₁₀) at port-4 of wide waveguide, and P_3 is the power at port-3; $P_{30}^{TE(TM)}$ is the power in TE₀₀ (TM₀₀) at port P_3 .

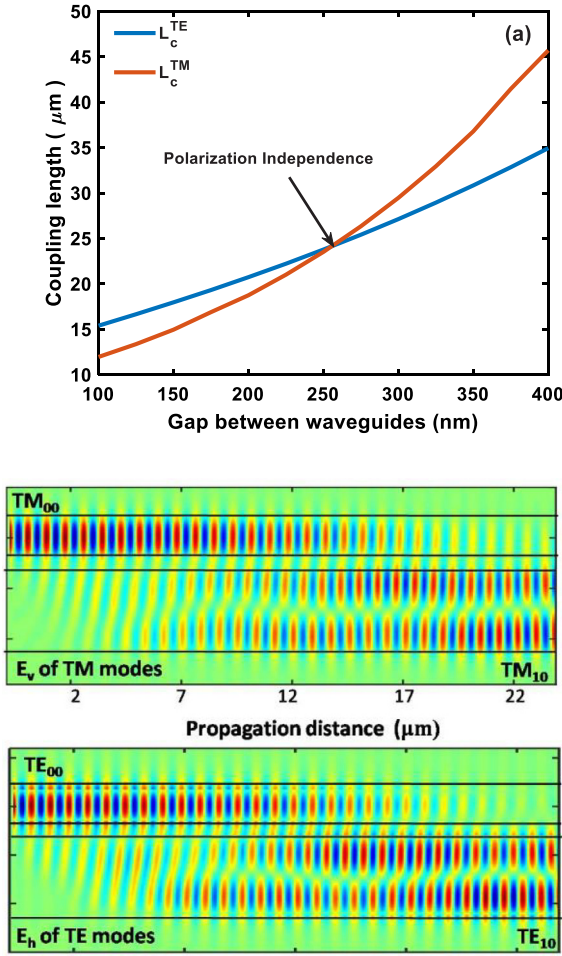


Fig. 7. (a) Coupling length as a function of separation between the waveguides (b) Dominant component (E_v) of TM mode (c) Dominant component (E_h) of TE mode along the propagation.

The analysis is extended to study fabrication tolerances to waveguide width, etch depth and sidewall angle deviations as shown in Figs. 8 and 9. MCE remains $> 90\%$ with width variation (Δw) of ± 30 nm. It is observed that MCE deviates from the trend and suddenly increases for TE modes at $\Delta w = +20$ nm. This happens because of the coupling length at this width becoming equal to the chosen length of straight waveguides (23.61 μm). Proposed design offers an etch depth tolerance (Δh) of ± 18 nm while maintaining MCE more than 90%. With the increase in etch depth deviation, MCE for TE modes falls off steeply due to the increase in phase mismatch between TE_{00} of narrow and TE_{10} mode of wide waveguide. For example, at $\Delta h = +20$ nm, the difference between the effective indices of these modes becomes 10.69×10^{-3} which is double than that at the actual design. Plots also show that TE and TM follow opposite trends for geometrical parameter deviations and our design is a trade-off to obtain best performance for both the polarizations. Tolerance to sidewall angle is shown in Fig. 9(a) and (b). Bandwidth of the demux for which crosstalk is ≤ -10 dB is found to be ~ 165 nm, as inferred from Fig. 9(d).

Demux using straight waveguides has been demonstrated as a proof of concept, while for practical implementation, the two straight waveguides have to be decoupled using bends. To test the demux performance, simple S-bends have been designed at the input and output ports. As the S-bend also has a straight section which adds to the coupling length, dimension of straight sections reduce to 11.5 μm . MCE was found to be 85% and 96% for TE and TM polarizations respectively.

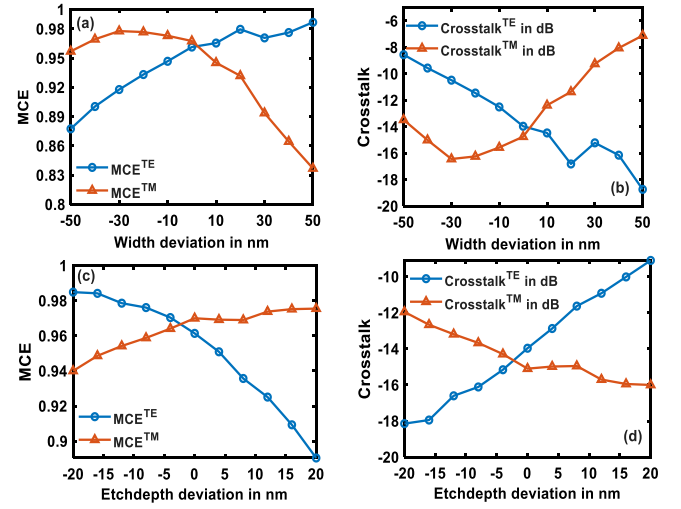


Fig. 8. Tolerance of MCE to (a) width variation (c) etch depth variation; Crosstalk as a function of (b) width variation (d) etch depth variation.

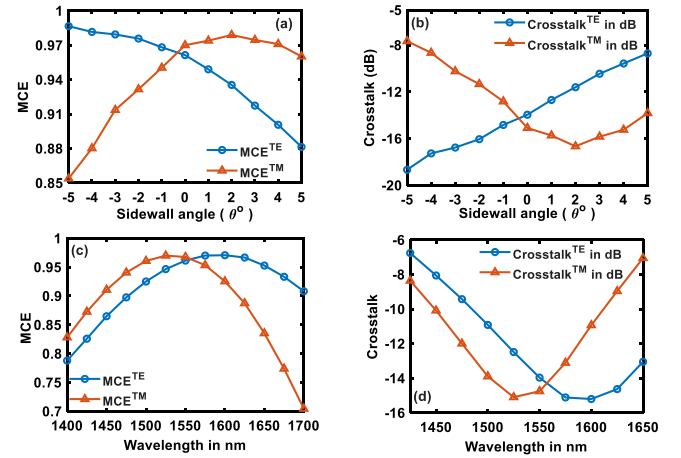


Fig. 9. Effect of sidewall angle variation on (a) MCE (b) Crosstalk. Variation of (c) MCE and (d) crosstalk as function of wavelength.

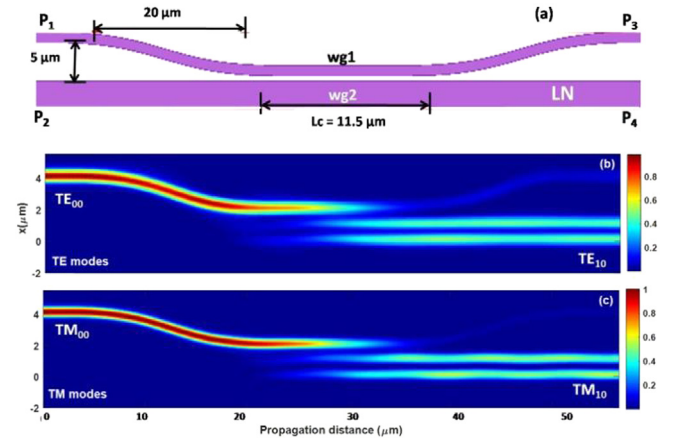


Fig. 10. (a) 2D view of demux with S-bends. Electric field distribution when (b) TE_{00} (c) TM_{00} is launched into the narrow waveguide.

The electric field distribution when TE_{00} and TM_{00} is launched into the narrow waveguide is shown in Fig. 10(b) and (c). Minimum bending radius for an acceptable bending loss in TE and TM modes

is different [27] and, the design optimization of bends in order to achieve best performance for both the orthogonal polarizations will be taken up as future work. Implementation of electro-optic effect would be an interesting feature to add, which facilitates the possibility of reconfigurable multimode directional couplers.

5. Conclusion

In conclusion, we have carried out systematic numerical analyses to show that anisotropy of LN can be utilized to tune the effective indices of qTE and qTM modes to achieve zero birefringence in dual-mode waveguides. We have demonstrated a birefringence which is zero to four decimal places for both the fundamental mode pair and the first order mode pair, which translates to a coherence length of about 38 mm when operated at a wavelength of 1550 nm. Even though the structure is tolerant to large fabrication errors in rib width and sidewall angle, its sensitivity to etch depth limits practical coherence length to about 1.4 mm. Proposed sample waveguide design offers a large fabrication tolerance of ± 400 nm for width, ± 8 nm for the etch depth, and $\pm 4^\circ$ for the sidewall angle which can achieve a very low birefringence of <0.001 . It will be interesting to enhance this design to achieve minimal birefringence in the pulsed regime as well by minimizing difference between group indices of modes.

Polarization-independent two-mode mux/demux has been designed using dual-mode ZBR waveguides. Coupling lengths for TE and TM polarizations are found to differ by just 0.26 μm . Proposed demux exhibits an MCE greater than 96% and a crosstalk of about -14 dB for both the polarizations. This device also exhibits good fabrication tolerance and has a bandwidth of 165 nm. Reported results pave the way for polarization-independent multimode devices that find use in mode/polarization division hybrid multiplexing, multi-analyte sensing applications and interferometric devices.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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